



Inverse Folding
(gRNAde)

Multi-round update
with adaptive α

Chosen

A G U G ... A C C

Rejected

A C A G ... U A C

Evaluation

Sequence AUCG
Secondary structure
Thermostability
Tertiary structure
SSTT Benchmark

Generating and constructing preference pairs

A G U G ... A C C

A C A G ... U A C

⋮

A C A G ... U A C

Generated sequences

Folding Model
(RhoFold+)

Thermostability
(ViennaRNA)

Margin condition

$\Delta\text{RMSD} > \alpha \text{ std}$

$\Delta\text{pLDDT} > \alpha \text{ std}$

$\text{RMSD} < \gamma$

$\text{pLDDT} > \kappa$

Filter condition

Chosen

A G U G ... A C C

Rejected

A C A G ... U A C

Preference pair construction